Distributed Systems for Conversational AI

From Rule-Based Chatbots to Distributed, Multimodal Intelligent Systems

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INTRODUCTION

What is Distributed Multimodal Conversational AI?

Multimodal Conversational AI enables natural dialogue across speech, text, and vision with Large Language Models as the reasoning core. Distributed architectures place components across edge devices, microservices, and cloud to achieve low latency, scalability, and improved privacy for real-world applications.

The integration of distributed computing is fundamental to deploying these systems at scale. By partitioning workloads across edge devices and microservices, modern conversational AI achieves fault tolerance and privacy-preserving processing — critical for production-grade applications.

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Evolution of Multimodal Conversational AI

Rule-Based Systems (ELIZA)

Pattern matching, rigid dialogues, domain-constrained. No learning capability.

Statistical & ML Models

RNN encoders, seq2seq pipelines. Early STT/NLP integration enabled data-driven responses.

Transformer Era

BERT and GPT improved coherence. Attention mechanisms enabled long-range context understanding.

LLM-Powered Agents

Function calling enables task execution and device control via natural language.

Distributed Multi-Agent Systems

Edge-cloud hybrids with open interoperability APIs for scalable, real-world deployment.

The arc moves from brittle rule systems to flexible, distributed agents that reason across modalities and act in the world.

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Base Research Papers Overview — Papers 1, 2, 3

Real-Time STT on Edge (Di Leo et al., MDPI, 2025)

Contribution: Edge-deployed STT prototype achieving sub-second latency using entirely local infrastructure

Key Technologies: WebRTC, WebSocket, Vosk ASR engine, Flask NLP correction microservice

Addresses: Cloud dependency, user privacy, and bandwidth limitations in real-time voice processing

State-of-the-Art in Conversational AI (Adewumi et al., MDPI, 2022)

Contribution: Survey of SoTA open-domain models — BlenderBot, Meena, LaMDA — mapping progress and persistent gaps

Key Technologies: Retrieval-based, generation-based, and hybrid dialogue; BLEU, perplexity, SSA evaluation metrics

Addresses: Bland and repetitive responses, figurative language failure, low-resource language support, gender bias in AI

Multi-Agent Interoperability (Gosmar et al., arXiv/OVON, 2024)

Contribution: OVON framework defining universal Natural Language APIs for loosely coupled multi-agent AI systems

Key Technologies: Conversation Envelope (JSON), Natural Language event APIs, Discovery & Manifest specification

Addresses: Fragmented AI ecosystems, vendor lock-in, and lack of interoperability across voice, text, and video agents

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Base Research Papers Overview — Papers 4, 5

LLM-Powered Agent for IoT Automation (Gallo et al., Springer, 2024)

Contribution: RuleBot++ — GPT-4 conversational agent enabling smart home automation through natural language

Key Technologies: GPT-4 function calling, trigger-action paradigm, Home Assistant, YAML validation, MongoDB

Addresses: IoT configuration complexity for non-expert users; converts spoken intent into executable automation

Impact: User study with 16 participants showed superior usability over traditional visual automation interfaces

Learning towards Conversational AI (Fu et al., Elsevier, 2022)

Contribution: Structured taxonomy of open-domain dialogue across retrieval, generation, and hybrid frameworks

Key Technologies: CNN, RNN, BERT, Variational Autoencoders (VAE), seq2seq encoder-decoder architectures

Addresses: Producing responses that are simultaneously informative, diverse, and stylistically controllable

Impact: Foundational reference for LLM training strategies and evaluation in multi-turn open-domain dialogue

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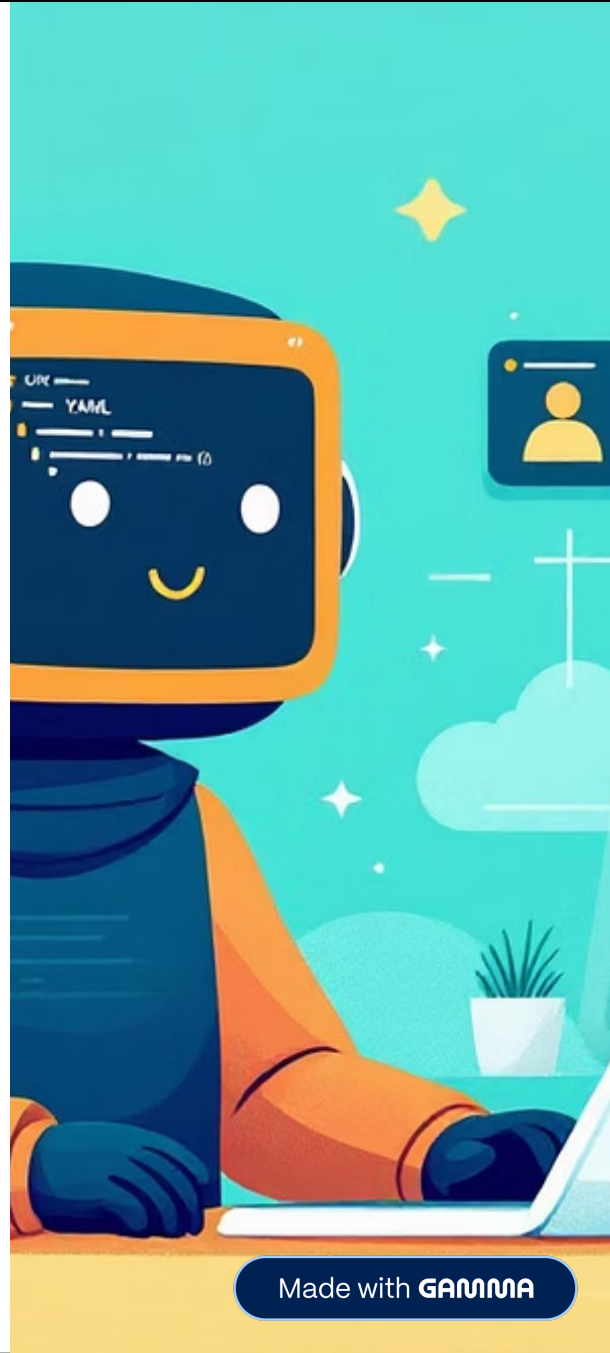
Current Trends — Edge & Interoperability

Edge-Deployed Inference

Shift from cloud to edge: sub-second STT, offline and privacy-preserving deployments (Di Leo et al., 2025). Reduces latency and bandwidth costs for real-time applications.

Open Multi-Agent Interoperability

OVON's Conversation Envelope enables cross-platform voice, text, and video exchange without vendor lock-in. Standardized protocols make agent handoffs more reliable.



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Current Trends — LLM Automation & Hybrid Dialogue

LLM-Powered Automation

GPT-4 function calling converts conversational intent into executable actions — device control, database queries, structured responses. Shifts user interaction from command-based to intent-based workflows. Demonstrated by RuleBot++ (Gallo et al., 2024), where non-expert users configured smart home devices entirely through natural language — no technical knowledge required.

Hybrid Dialogue Architectures

Retrieval ensures factual accuracy; generation preserves linguistic diversity. Combining both reduces hallucinations while keeping responses natural and context-aware. This hybrid paradigm — studied across Fu et al. and Adewumi et al. — is now the dominant architecture in production conversational AI systems, balancing knowledge grounding with fluent, human-like output..

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Future Enhancements in AI

Low-Latency Inference

Sub-50ms responses via edge accelerators, model quantization, and optimized runtimes.

Federated AI

On-device training avoids centralizing sensitive data; supports personalization and privacy.

Universal Interoperability

Extending OVON Conversation Envelope to natively support audio, video, and semantic metadata across platforms.

Multimodal Fusion

Unified reasoning across voice, vision, and text using cross-modal attention for coherent contextual responses.

Ethical & Inclusive AI

Accent-aware models, multilingual support, and gender-neutral design to reduce bias and increase access.

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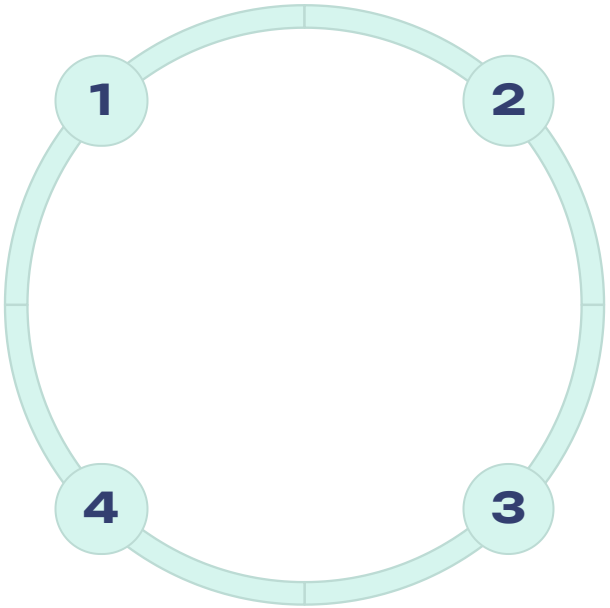
Real-World Applications

Healthcare

On-device clinical documentation, patient triage, and privacy-preserving assistants in sensitive settings.

Accessibility

Offline captioning, TTS for visually impaired users, and multilingual interfaces without internet dependency.



Smart Environments & IoT

Natural language device control and trigger-action automation for non-expert users (Gallo et al., 2024).

Enterprise & Customer Service

Multi-agent networks route queries across specialized bots with seamless handoffs and context persistence.

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Conclusion — The Path Forward

Surveyed work highlights a clear convergence toward edge inference, open interoperability (OVON), and LLM-driven automation. Core technologies — offline ASR, GPT-4 function calling, and Conversation Envelope APIs — enable systems that are fast, private, and platform-agnostic.

The five papers collectively define the architectural vocabulary of next-generation conversational systems — from lightweight edge STT to multi-agent frameworks capable of coordinating across voice, text, and video channels.

The path forward demands convergence on federated learning, universal agent interoperability standards, and multimodal LLM fusion — making distributed conversational AI the foundational layer of human-machine interaction.

References: Di Leo et al. (2025); Adewumi et al. (2022); Gosmar et al. (OVON, 2024); Gallo et al. (2024); Fu et al. (2022).

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